Euclid's Elements — Book I
Proposition 29

Formal Reconstruction — Technical Documentation

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A straight line falling on parallel straight lines makes the alternate angles equal to one another, the exterior angle equal to the interior and opposite angle, and the interior angles on the same side equal to two right angles.

1 Introduction

Proposition I.29 is the first proposition in Book I whose proof genuinely uses the Euclidean parallel axiom.

The preceding Propositions I.27 and I.28 establish implications of the form

$$\text{angle condition} \longrightarrow \text{parallel lines.}$$

Those implications belong to neutral geometry. Proposition I.29 reverses the direction:

$$\text{parallel lines} \longrightarrow \text{angle relations.}$$

This is the point at which Hilbert's Group IV enters the reconstruction.

The proposition has three conclusions. If a transversal cuts two parallel straight lines, then:

$$\angle AGH \cong \angle GHD,$$

the alternate interior angles are congruent;

$$\angle EGB \cong \angle GHD,$$

the exterior angle is congruent to the interior and opposite angle on the same side; and the same-side interior angles

$$\angle BGH \qquad \text{and} \qquad \angle GHD$$

are together equal to two right angles.

Only the first conclusion contains the genuinely Euclidean step. The other two are consequences of the first by vertical-angle and supplementary-angle geometry.

2 Classical Configuration

Let the straight line EF meet the parallel straight lines AB and CD at G and H, respectively:

$$A - G - B, \qquad C - H - D, \qquad E - G - H - F.$$

Assume

$$AB \parallel CD.$$

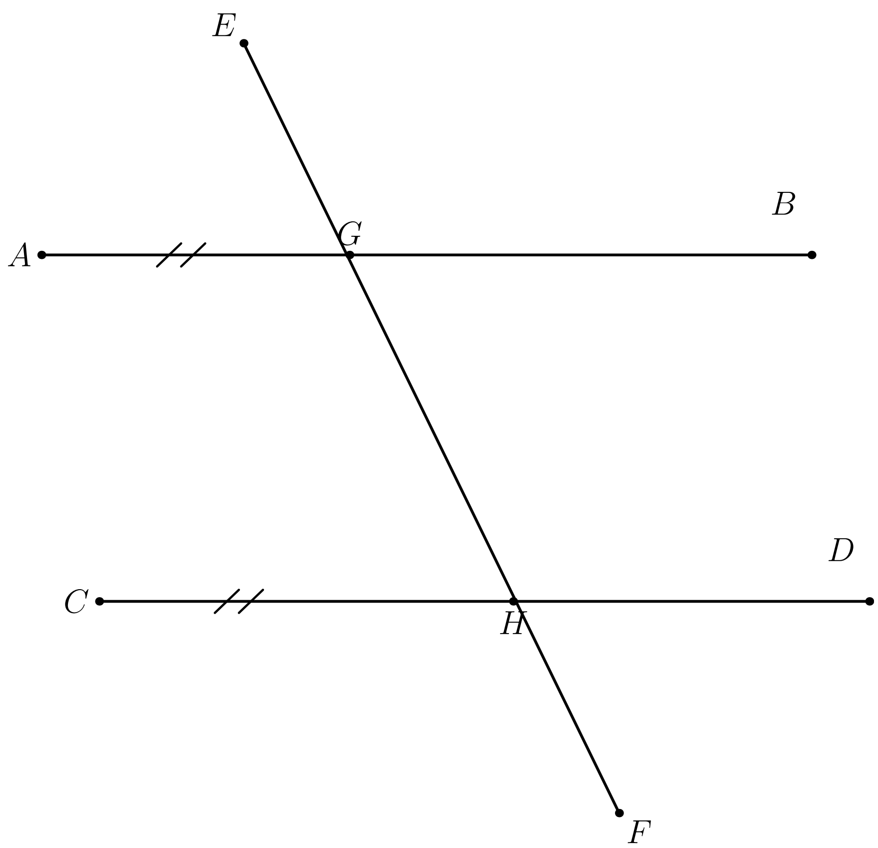

Figure 1: The transversal configuration of Proposition I.29. The straight lines AB and CD are parallel.

3 First Conclusion: Alternate Interior Angles

The principal assertion is

$$\angle AGH \cong \angle GHD.$$

This is the converse direction of Proposition I.27.

In the Hilbert reconstruction the corresponding general theorem is the Euclidean direction of Hilbert's Theorem 30. The proof constructs through the second intersection point a line making the required alternate angle. The neutral alternate-angle criterion shows that this constructed line is parallel to the first given line. Hilbert's parallel axiom then identifies the constructed line with the already given parallel line.

Structurally, the first part is exactly the construction mechanism isolated later in Proposition I.31: copy an arbitrary angle at the external point and use I.27 to obtain a parallel. No Euclidean parallel axiom is involved in this existence step.

Schematically,

$$\begin{gathered} AB \parallel CD \\ \downarrow \\ \text{construct through } H \text{ a line with the required alternate angle} \\ \downarrow \\ \text{neutral alternate-angle criterion} \\ \downarrow \\ \text{constructed line is parallel to } AB \\ \downarrow \text{ Hilbert Group IV} \\ \text{constructed line} = CD \\ \downarrow \\ \angle AGH \cong \angle GHD. \end{gathered}$$

4 Second Conclusion: Corresponding Angles

Once

$$\angle AGH \cong \angle GHD$$

has been established, the second conclusion is neutral.

Because $A - G - B$ and $E - G - H$, the angles

$$\angle AGH \qquad \text{and} \qquad \angle EGB$$

are vertical angles. Proposition I.15 therefore gives

$$\angle EGB \cong \angle AGH.$$

Combining this with the alternate-angle conclusion yields

$$\angle EGB \cong \angle GHD.$$

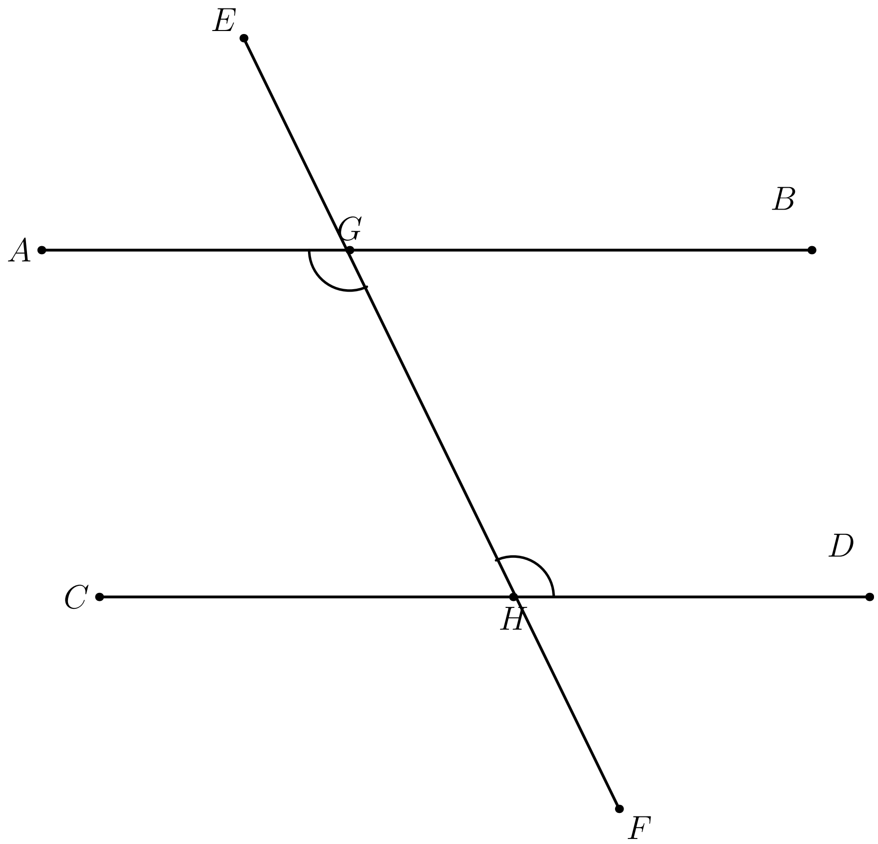

Figure 2: The Euclidean core of Proposition I.29: parallelism implies congruence of the alternate interior angles.

Thus the second branch has the dependency pattern

$$\text{I.29 alternate} \longrightarrow \text{I.15 vertical angles} \longrightarrow \text{corresponding-angle congruence.}$$

5 Third Conclusion: Same-Side Interior Angles

The third conclusion states that

$$\angle BGH \qquad \text{and} \qquad \angle GHD$$

are together equal to two right angles.

The reconstruction does not introduce numerical angle measure or a binary operation of angle addition. Instead it uses the synthetic predicate

`HilbertAnglesEqualTwoRightAnglesWithSupplement`

already placed in the Hilbert interface.

Because

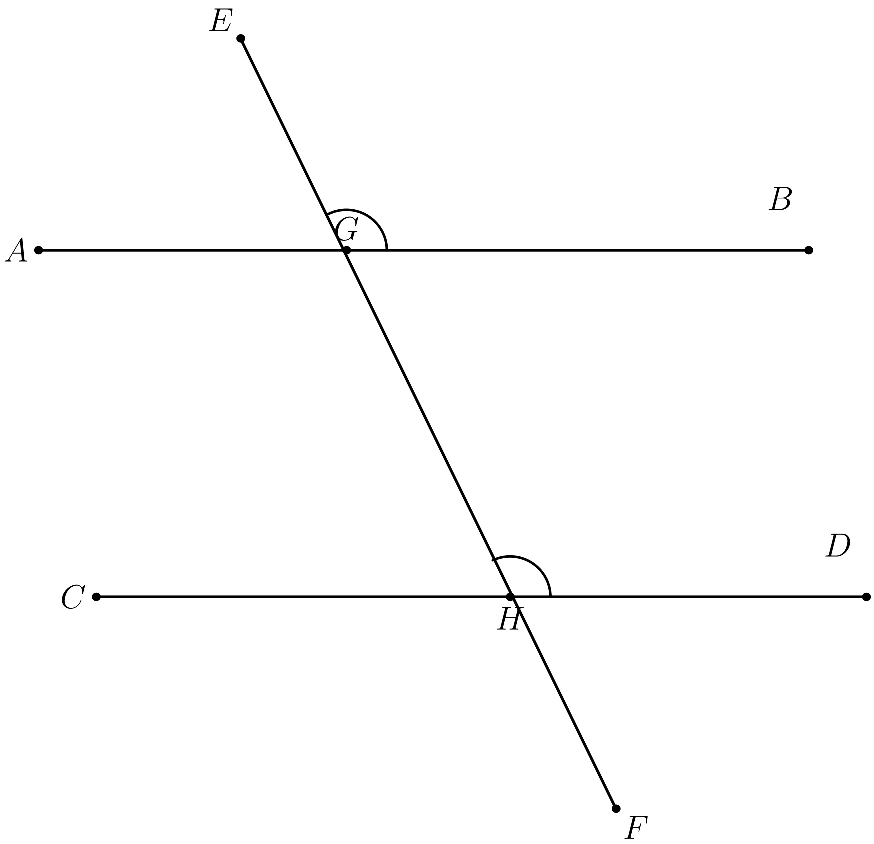

Figure 3: The corresponding-angle conclusion. The exterior angle $\angle EGB$ is vertical to $\angle AGH$, which is already congruent to $\angle GHD$.

$$A - G - B,$$

the rays GA and GB are opposite. Hence $\angle AGH$ is the supplementary angle associated with $\angle BGH$.

From the first conclusion,

$$\angle AGH \cong \angle GHD.$$

Therefore the same-side pair satisfies the synthetic condition expressing that it is together equal to two right angles.

The dependency is

$$\begin{gathered} \angle AGH \cong \angle GHD \\ A - G - B \\ \downarrow \\ \text{supplementary configuration} \\ \downarrow \\ \angle BGH + \angle GHD = 2R \quad \text{(synthetically)}. \end{gathered}$$

6 Lean Formalization

The final Lean development contains five theorems.

The interface-level form is

```
euclid_proposition_29
```

and is a direct wrapper around

```
equal_angles_from_parallel
```

which is the Euclidean direction of Hilbert's Theorem 30.

The explicit transversal form is

```
euclid_proposition_29_transversal
```

with the conclusion

$$\angle AGH \cong \angle GHD.$$

Its proof first transports the global parallel relation

$$AB \parallel CD$$

to the local lines determined by the actual angle arms,

$$GA \parallel HD.$$

A point M is then chosen with

$$G - M - H.$$

The line-level Hilbert theorem gives

$$\angle MGA \cong \angle MHD.$$

Angle reversal and same-ray transport along the transversal convert this into

$$\angle AGH \cong \angle GHD.$$

The second theorem,

```
euclid_proposition_29_corresponding
```

combines the alternate-angle theorem with `VerticalAngles` to prove

$$\angle EGB \cong \angle GHD.$$

The third theorem,

```
euclid_proposition_29_same_side
```

combines the alternate-angle theorem with the supplementary relation $A - G - B$ and returns

```
HilbertAnglesEqualTwoRightAnglesWithSupplement
```

for the same-side interior angles.

Finally,

```
euclid_proposition_29_all
```

packages the three conclusions in one theorem.

7 Proof Architecture

The formal dependency structure is

$$\begin{array}{c} AB \parallel CD \\ \downarrow \\ \texttt{hilbert\_alternate\_angles\_of\_parallel\_oppositeSide\_lines} \\ \downarrow \text{ Hilbert Group IV} \\ \angle AGH \cong \angle GHD \end{array}$$

and then

$$\begin{array}{ccc} & \angle AGH \cong \angle GHD & \\ \swarrow & & \searrow \\ \text{vertical angles} & & \text{supplementary configuration} \\ \downarrow & & \downarrow \\ \angle EGB \cong \angle GHD & & \angle BGH + \angle GHD = 2R. \end{array}$$

Thus the Euclidean axiom is needed only once. After the alternate-angle conclusion has been obtained, the remaining two conclusions are neutral.

8 The Logical Boundary at I.29

Propositions I.27–I.29 form a particularly clear logical block.

$$
\begin{aligned}
\text{I.27} &: \text{ equal alternate angles} \longrightarrow \text{parallel lines,}\\
\text{I.28} &: \text{ other angle conditions} \longrightarrow \text{parallel lines,}\\
\text{I.29} &: \text{ parallel lines} \longrightarrow \text{angle relations.}
\end{aligned}
$$

The first two implications are neutral. The third is Euclidean.

This distinction is visible directly in the Lean typeclasses: I.27 and I.28 do not require the Euclidean plane structure, while I.29 uses

```
[HilbertEuclideanPlane Geo]
```

at the theorem that converts parallelism into alternate-angle congruence.

9 Existence and Uniqueness Behind I.29

The first conclusion of I.29 admits a useful structural decomposition.

The neutral part of the argument constructs through H a line having the required direction. Angle construction supplies the copied angle, and I.27 proves that the constructed line is parallel to AB. This is the same existence mechanism that appears explicitly in Proposition I.31.

At this stage neutral geometry gives only

$$\text{there exists a line through } H \text{ parallel to } AB.$$

But the hypotheses of I.29 already contain another line through H, namely CD, with

$$AB \parallel CD.$$

To conclude that the newly constructed line is actually CD, one needs uniqueness of the parallel through H. This is precisely the point at which Hilbert's Group IV, equivalently the Playfair-type uniqueness principle used in the reconstruction, enters.

Thus the logical core of the first conclusion can be displayed as

$$
\begin{array}{c}
\text{angle construction}\\
\downarrow\\
\text{equal alternate angles}\\
\downarrow\\
\text{I.27}\\
\downarrow\\
\text{a constructed parallel through } H\\
\downarrow \;\; \text{Group IV / parallel uniqueness}\\
\text{the constructed line is } CD\\
\downarrow\\
\text{the required angle congruence of I.29.}
\end{array}
$$

This makes precise the observation that the Euclidean axiom is needed only once. It is not needed to construct a parallel direction; it is needed to identify the constructed parallel with the parallel line already present in the hypotheses.

Wilkins' discussion of the parallel propositions highlights this existence–uniqueness separation. Read retrospectively after I.31, the mechanism can be summarized schematically as

$$\text{I.31-type construction} + \text{I.27} + \text{Playfair uniqueness} \Longrightarrow \text{the angle conclusion of I.29.}$$

This is a structural analysis of I.29, not a claim that Euclid's original proof of I.29 depends on the later Proposition I.31.

10 Conclusion

Proposition I.29 is the first genuine Euclidean-parallelism theorem in the Book I reconstruction.

Its essential mathematical content is the single implication

$$AB \parallel CD \longrightarrow \angle AGH \cong \angle GHD.$$

That implication is Hilbert's Euclidean direction of the alternate-angle theorem and depends on Group IV.

The remaining statements of Euclid's proposition are consequences of this result. Corresponding angles are obtained through vertical-angle congruence, and the same-side two-right-angles condition is represented synthetically by a supplementary configuration.

The resulting formalization therefore isolates the exact point at which the parallel axiom enters and avoids attributing additional Euclidean strength to the two corollary branches.